II. Airframe

Subject E. Landing Gear Systems

Objective	*The following knowledge, risk management, and skill elements are required for aircraft landing gear systems.*
Knowledge	*The applicant demonstrates understanding of:*
AM.II.E.K1	Fixed and retractable landing gear systems.
AM.II.E.K2	Fixed and retractable landing gear components.
AM.II.E.K3	Landing gear strut servicing/lubrication.
AM.II.E.K4	Inspection of bungee and spring steel landing gear systems.
AM.II.E.K5	Steering systems.
AM.II.E.K6	Landing gear position and warning system inspection, check, and servicing.
AM.II.E.K7	Brake assembly servicing and inspection.
AM.II.E.K8	Anti-skid system components and operation.
AM.II.E.K9	Wheel, brake, and tire construction.
AM.II.E.K10	Tire storage, care, and servicing.
AM.II.E.K11	Landing gear and tire and wheel safety and inspection.
AM.II.E.K12	Brake actuating systems.
AM.II.E.K13	Alternative landing gear systems (e.g., skis, floats).
Risk Management	*The applicant demonstrates the ability to identify, assess, and mitigate risks associated with:*
AM.II.E.R1	Landing gear and tire and wheel practices/precautions.
AM.II.E.R2	Use of aircraft jacks.
AM.II.E.R3	High pressure fluids and gases.
AM.II.E.R4	Storage and handling of hydraulic fluids.
AM.II.E.R5	High pressure strut or system disassembly.
AM.II.E.R6	Operation of retractable landing gear systems around personnel.
Skills	*The applicant demonstrates the ability to:*
AM.II.E.S1	Inspect and service landing gear.
AM.II.E.S2	Inspect, check, and service an anti-skid system.
AM.II.E.S3	Locate and explain procedures for checking operation of an anti-skid warning system.
AM.II.E.S4	Locate and explain troubleshooting procedures for an anti-skid system.
AM.II.E.S5	Jack aircraft.
AM.II.E.S6	Troubleshoot a landing gear retraction check.
AM.II.E.S7	Inspect wheels, brakes, bearings, and tires.
AM.II.E.S8	Remove and replace brake lining(s).
AM.II.E.S9	Service landing gear air/oil shock strut.
AM.II.E.S10	Bleed air from a hydraulic brake system.
AM.II.E.S11	Troubleshoot hydraulic brake systems.
AM.II.E.S12	Remove, inspect, and install a wheel brake assembly.

AM.II.E.S13	Inspect a tire for defects.
AM.II.E.S14	Locate tire storage practices.
AM.II.E.S15	Replace air/oil shock strut air valve.
AM.II.E.S16	Troubleshoot an air/oil shock strut.
AM.II.E.S17	Service a nose-wheel shimmy damper.
AM.II.E.S18	Inspect nose-wheel steering system for proper adjustment.
AM.II.E.S19	Locate and explain the process for checking landing gear alignment.
AM.II.E.S20	Replace master brake cylinder packing seals.
AM.II.E.S21	Troubleshoot aircraft steering system.
AM.II.E.S22	Identify landing gear position and warning system components.
AM.II.E.S23	Troubleshoot landing gear position and warning systems.
AM.II.E.S24	Inspect and repair landing gear position indicating system.
AM.II.E.S25	Adjust the operation of a landing gear warning system.
AM.II.E.S26	Remove, install, and adjust a landing gear down-lock switch.
AM.II.E.S27	Inspect a brake for serviceability.
AM.II.E.S28	Troubleshoot nose-wheel shimmy.
AM.II.E.S29	Inspect tube landing gear for damage.